

# Adam W. Lester, PhD

## Postdoctoral Research Fellow

School of Biomedical Engineering  
University of British Columbia, NC4 Lab  
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[Portfolio](#) • [GitHub](#) • [Google Scholar](#)  
[LinkedIn](#) • [Detailed CV](#)

## About

Neuroscientist and biomedical engineering postdoc with 15 years of basic and clinical research and 12 years of end-to-end research instrumentation development, including 10+ instruments deployed and in active use.

## Education

PhD Neuroscience University of Arizona  
BA Philosophy UC Santa Cruz

## Core Competencies

- Technical & Project Leadership
- Cross-functional Leadership
- Industry-Research Collaboration
- System Architecture
- Iterative Design & Validation
- Basic & Clinical Research Methods

## Technical Skills

### Languages & Platforms

- Python, C/C++, C#, MATLAB, R
- ROS, PlatformIO, Linux, Git/GitHub

### Control & Robotics

- Embedded systems (Pi, AVR, ARM)
- Closed-loop & real-time control
- Motion control (stepper, servo, GRBL)
- I/O (I<sup>2</sup>C, SPI, UART, TTL, EtherCAT)

### Design & Fabrication

- Mechanical CAD (SolidWorks)
- DFM, BOMs, QC, jigs & fixtures, ECAD
- Wiring harnesses, soldering/rework
- Additive/subtractive manufacturing

### Analytical Methods

- Data integration & standardization
- State-space modeling
- Computational modeling
- Signal processing
- Data visualization & statistics

**Publications:** see [CV](#) for selected list

## Selected Experience

- 2021 – **Postdoctoral Research Fellow** University of British Columbia  
Present • Led end-to-end design and deployment of 2 NC4 Lab open-source behavioral platforms ([Omniroute](#), [NC4touch](#)) and 3 subsystems, used collectively across 6 studies and 2 UBC labs
- Published 2 first-author methods: 1 in press (*eNeuro*) and 1 preprint
- Designed and co-led a human VR clinical study (completed, article in preparation) and a rodent navigation-network study (ongoing)
- Conceived, wrote, and won 2 competitive project grants
- Co-founding an early-stage UBC spinout with cross-department backing, open-sourcing custom instruments from UBC partner labs
- 2019 – **Postdoctoral Research Associate** University of Arizona  
2021 • Designed, led, and published an aging navigation study (*Behav Neurosci*, 2022) and wrote a funded NIA R01 renewal Specific Aim
- Designed a dual-site electrode implant used in 2 studies
- Led lab-side validation for 2 industry collaborations ([NeuraLynx](#), TBSI), developing wireless neural-recording from beta to release
- Supervised a 4-student engineering capstone through to release
- 2013 – **Graduate Research Associate** University of Arizona  
2019 • Architected, built, and deployed the ICR Arena behavioral platform with first-author published methods (*J Neurosci Methods*, 2020)
- Contributed to 4 co-authored studies on neural-behavioral mechanisms in navigation, learning, and decision making
- Stood up lab's UA and UC Davis rat and primate electrophysiology
- 2011 – **Computational Modeling Intern** HRL Laboratories, LLC  
2012 • Built a temporal-lobe gating ANN for the IARPA [ICArUS](#) human-sense-making program, and published a first-author *IJCNN* paper

## Selected Projects

- Integrated closed-loop research systems, with portfolio links*
- Omniroute** 90x90 cm rat maze with 60 addressable gates ([NC4gate](#)), projected cues, 3D tracking, [XY feeder gantry](#), and ROS control  
End-to-end lead: CAD (SolidWorks), ROS (Python, C++), GUI (Python), firmware (C/C++); led build, PCB dev, and validation teams
- NC4touch** 3-touchscreen chamber for mice and rats with automated reward, overhead video, and multi-chamber Raspberry Pi-based control  
Co-architect & hardware lead: CAD (SolidWorks), hardware integration, firmware; led build and embedded systems teams
- ICR Arena** 1.4 m augmented-reality rat arena with 360° projected cues, video tracking, and [mobile reward delivery](#)  
End-to-end lead: CAD (SolidWorks), fabrication, control stack/GUI (MATLAB, C#, C/C++), led build and validation teams

## Leadership & Service

- **Founding Product Lead:** MAPS Canada, Research Database (dev team of 4)
- **Guest lecturer:** BMEG 400F Biomedical Robotics; NRSC 315 Neuro Methods
- **Organizer:** NRSC 595B Theoretical Neuro JC; Ephys Summer Workshop
- **Open-Source:** 3 full releases with build docs (OSF) + code (GitHub)
- **Mentorship:** 17 trainees (UBC); 10 (UA); supervised 3 Honors theses